R2: Design Draft

EEL4907 CpE Design 1

Hardened RISC-V Exam Computer

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1 Introduction

1.1 Purpose / Need

The project aims to create a **plug-and-play secure exam computer** built around a RISC-V board. The motivation comes from the growing problem of AI-assisted cheating, especially in coding exams, and the limitations of current solutions such as Honorlock, which require intrusive monitoring of student laptops.

Instead of securing arbitrary student machines, we distribute **dedicated exam devices** during tests. Each device boots into a locked-down Linux environment and launches Safe Exam Browser (SEB) [1] so students take exams through Canvas. Because the hardware and software stack are controlled, tampering is much harder while avoiding the privacy concerns of remote proctoring tools. The system is needed by institutions and instructors who want exam integrity without surveilling students' personal devices.

1.2 Domain & Prior Work

The project sits at the intersection of secure boot, locked-down computing environments, and academic assessment. Prior work includes commercial proctoring (e.g., Honorlock, Respondus) and locked browsers (Safe Exam Browser). Our approach differs by controlling the entire device rather than layering software on untrusted student hardware.

Current status / prior work: This project was previously advanced by Nicholas DeSanctis. The RISC-V board already boots into a full Linux OS. Other students (Shane and Wilson) are integrating SEB with Canvas; our work will connect the RISC-V system to that pipeline.

1.3 Impact & Risk Assessment

The proposed hardened RISC-V exam computer addresses critical challenges in modern academic assessment, balancing security requirements with ethical and environmental considerations.

Social & Cultural: The primary social impact is the restoration of pedagogical integrity in an era where Large Language Models (LLMs) and generative AI can be deployed covertly to bypass traditional assessment. By hardening the testing environment, the project reinstates the incentive for students to master course material rather than relying on external automation. Culturally, this maintains the prestige of the University of Florida degree, ensuring that graduation signifies genuine competency and providing a level playing field for all students.

Economic: The current status quo for exam integrity relies on Software-as-a-Service (SaaS) models like Honorlock or Respondus. These solutions involve high operational expenditure, often reaching six-figure annual fees due to per-student, per-exam pricing structures. In contrast, this project represents a shift toward capital expenditure while the upfront hardware cost is higher, it eliminates recurring licensing fees. Additionally, maintaining a university-owned fleet creates technical student-employment opportunities, providing hands-on experience in open-source maintenance/development and hardware maintenance.

Ethical: Modern proctoring software often requires intrusive access to student-owned hardware and personal environments, raising significant privacy concerns. This project addresses these ethical dilemmas through a design-for-privacy approach, shifting the security burden from the student's personal device to university-owned hardware. This ensures that no invasive monitoring software—which can often act as a rootkit—is ever installed on a student's personal property.

Environmental & Global: Utilizing the RISC-V architecture promotes an open-standard Instruction Set Architecture (ISA), reducing global reliance on proprietary silicon. From an environmental standpoint, these boards are high-efficiency, low-power devices (typically <15W), significantly reducing the aggregate carbon footprint compared to a classroom of high-performance consumer laptops. The use of 3D-printed enclosures further allows for localized manufacturing and the use of recyclable materials like PETG or PLA.

Risk Assessment & Limitations: The primary technical risk is the "arms race" of hardware exploits. Since students have physical access to the device, sophisticated actors could attempt to interface with the hardware via GPIO pins or physical keyloggers. Furthermore, as a custom solution, there is a risk of hardware failure during an active exam. To mitigate these risks, the design incorporates a secure boot chain to detect system tampering and a deployment strategy that replaces faulty units immediately without loss of student exam progress.

2 Statement of Work

Primary Deliverable: The development of a RISC-V native, SEB-compatible kiosk environment. Because official SEB support is restricted to Windows, macOS, and iOS, this project will implement a **Linux SEB-Wrapper**. This wrapper will utilize a Wayland-based Kiosk compositor to replicate the Kiosk application lockdown features and a custom browser instance to handle the Browser Exam Key (BEK) and Configuration Security Key (CSK) handshakes required by Canvas.

Performance Specifications:

- **SoC:** SpacemiT K1 (8-core RISC-V) with hardware-accelerated AES/SHA for configuration decryption.
- **Lockdown Efficacy:** System must disable all TTY switching (Ctrl+Alt+Fx), terminate non-whitelisted processes, and prevent escape via standard window management shortcuts.
- **Integrity:** Use RISC-V PMP (Physical Memory Protection) to isolate the SEB configuration keys from user-space memory.

2.1 Hardware Candidate Analysis

The project evaluates two primary RISC-V Single Board Computers (SBCs) for the exam device. The selection is driven by the need for hardware-level security features and sufficient memory to handle the Chromium-based Safe Exam Browser (SEB) environment.

Table 1: Technical Comparison of Hardware Candidates

Feature	Banana Pi BPI-F3	Orange Pi RV
SoC	SpacemiT K1 (8-core)	StarFive JH7110 (4-core)
RAM	4GB LPDDR4X	2GB LPDDR4
Boot Flow	Vendor-specific SPL/SBI	Vendor-specific SPL/SBI

Design Constraint: Platform Portability
The Orange Pi RV's 2GB RAM risks OOM failures during Canvas exams. Due to non-standardized RISC-V firmware and OpenSBI, porting constitutes an architectural pivot exceeding the semester's scope. Consequently, the Banana Pi is the tentative sole deployment target.

2.2 High-Level Task Breakdown

ID	Task Description
SW-01*	Port SEB configuration parsing logic (XML/PList) to C++/Python for Linux.
SW-02	Implement SEB-to-LMS Handshake (BEK/CSK hashing) to satisfy Canvas security.
UI-01	Configure Cage (Wayland compositor) for single-application kiosk mode.
SEC-01	Implement Secure Boot and read-only eMMC partitioning on the BPI-F3.

Table 2: Project Milestones for SEB-Linux Implementation (: First Priority)*

3 Deliverable Artifacts

3.1 Hardware

Description: A simple physical exam device for easy deployment (e.g., 3D-printed enclosure mounting the RISC-V board to the VESA mount on the back of a monitor, creating a compact "all-in-one" exam computer with only power, keyboard, and mouse).

Requirements & Specifications:

- **Memory:** A minimum of 4GB RAM is preferred to ensure stability. While 2GB models (e.g., Orange Pi RV) are available, they risk session instability when running modern web-based Learning Management Systems.
- **Firmware:** The device utilizes a custom-built Linux image. Due to the lack of architectural standardization in the RISC-V SBC ecosystem, the bootloader (U-Boot) and firmware stack are specifically tailored to the Banana Pi BPI-F3's vendor-defined boot flow.
- **Security:** Utilization of RISC-V Physical Memory Protection (PMP) to isolate the M-mode (Machine) security environment from the U-mode (User) browser environment.

Accessibility, Usability, & Maintenance: The hardware utilizes a standard VESA mount to ensure compatibility with existing university laboratory infrastructure. For maintenance, the enclosure is designed for tool-less or single-fastener access to allow for rapid replacement of the RISC-V board or storage media in the event of hardware failure. Usability is prioritized through a simplified I/O layout, reducing the potential for student error if connecting peripherals.

3.2 Software

Description: Hardened Linux image, secure boot components, and integration with Safe Exam Browser and Canvas.

Dissemination Plan: All produced software will be open sourced, barring components that must remain confidential to maintain the hardening integrity of the system. By design, the software will be packaged and integrated with the OS of the device and will require no additional building, installation, or configuration by the end-user.

Accessibility, Usability, & Maintenance: The software environment is designed to be self-healing; since the eMMC is mounted as a read-only filesystem, any accidental or malicious changes are wiped upon a power cycle. For accessibility, the kiosk browser will support standard web accessibility features, including screen reader compatibility via the ARIA standard and keyboard-only navigation, ensuring compliance with ADA requirements for testing environments.

3.3 Documentation

Description: Manuals, specifications, and reports (including this design draft and subsequent deliverables).

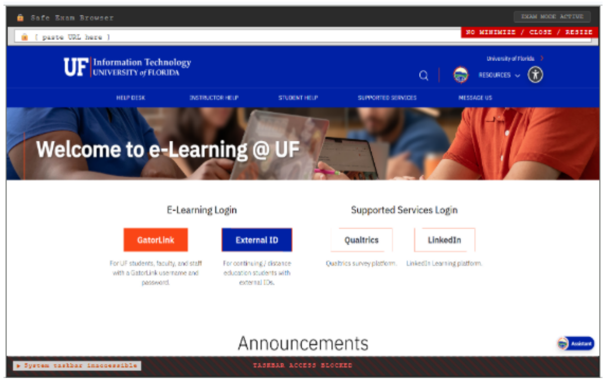
Dissemination Plan: All produced documentation will be open sourced alongside the project software. Additionally, documentation will be provided for use and procedures at each level of authority. That is, documentation will be provided for those who intend to build the project, professors who aim to use the device, and students requiring instructions on proper operation.

Accessibility, Usability, & Maintenance: All documentation will be provided in searchable, screen-reader-friendly PDF and Markdown formats to ensure accessibility for all users. Maintenance will be managed via a hosted documentation platform (e.g., ReadTheDocs or a GitHub Wiki), allowing for real-time updates as the RISC-V ecosystem or Canvas API evolves. Versioned manuals will ensure that documentation remains synchronized with the specific hardware revisions deployed in the field.

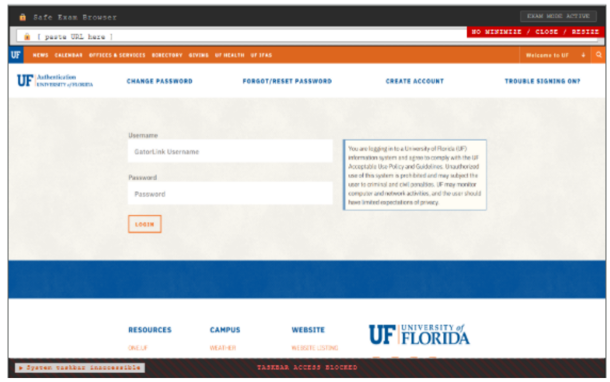
4 Mockups

4.1 Interfaces

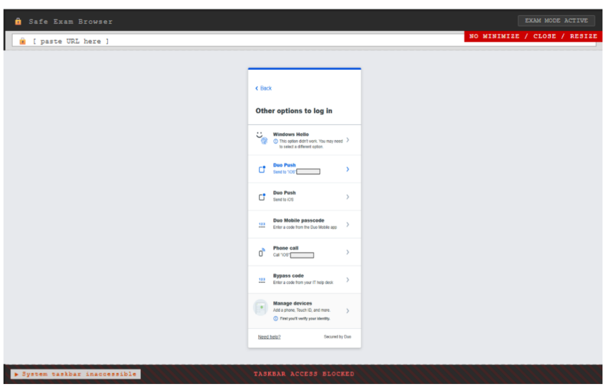
The following mockups illustrate each software screen a student will encounter during an exam session on the hardened RISC-V exam computer.



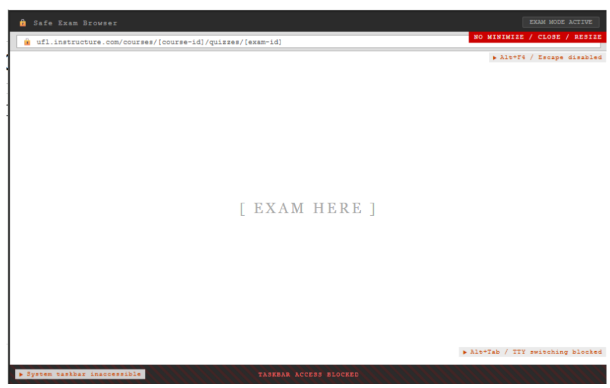
(a) UF e-Learning Login



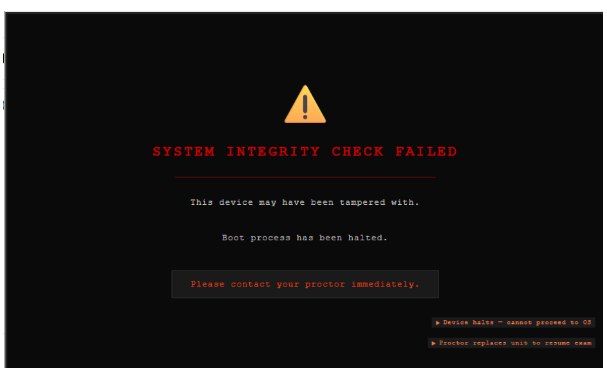
(b) Gatorlink Authentication



(c) Duo Two-Factor Auth



(d) Canvas Exam Session



(e) System Integrity Check Failure

Figure 1: Interface mockups: software screens within the SEB kiosk (a–e).

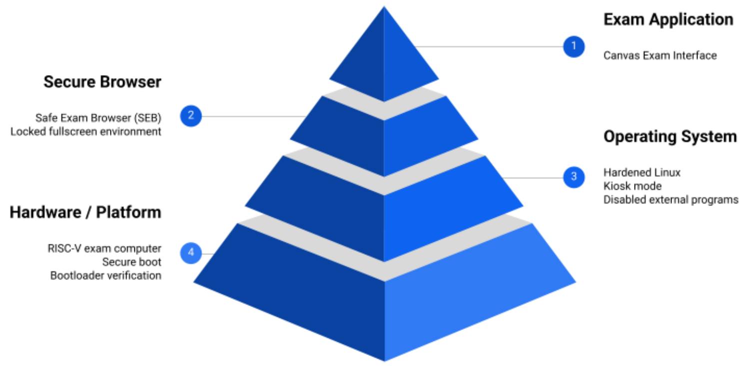


Figure 2: Layered architecture of the system showing hardware, OS, browser, and application layers

4.2 Systems

4.3 Networking

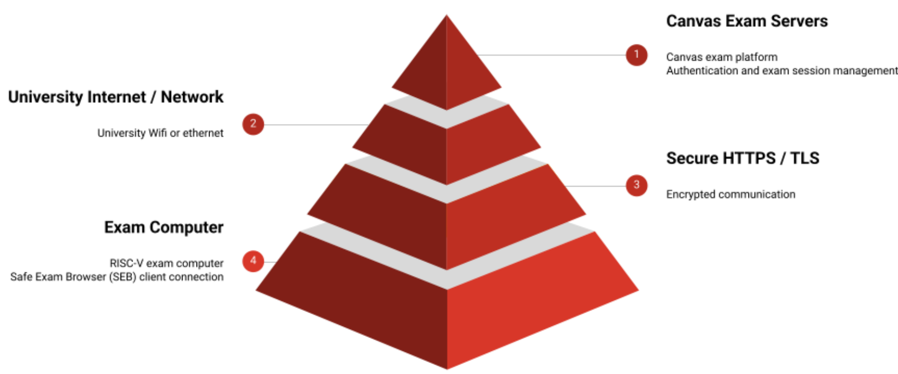


Figure 3: Network layers showing secure communication from Canvas servers to the exam device

4.4 Storyboards

The following storyboard illustrates the complete software flow a student experiences from device power-on to exam completion.

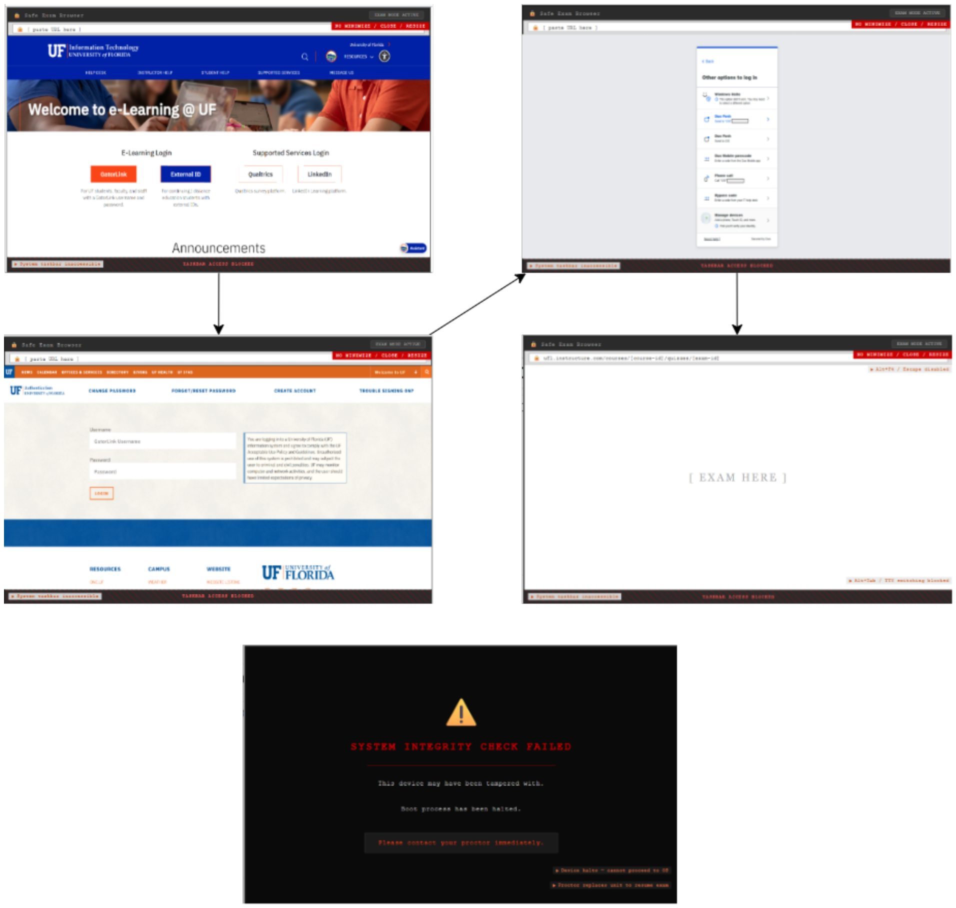


Figure 4: Student Workflow Storyboard: SEB Launch through Canvas Exam Completion

4.5 Draft Schematics

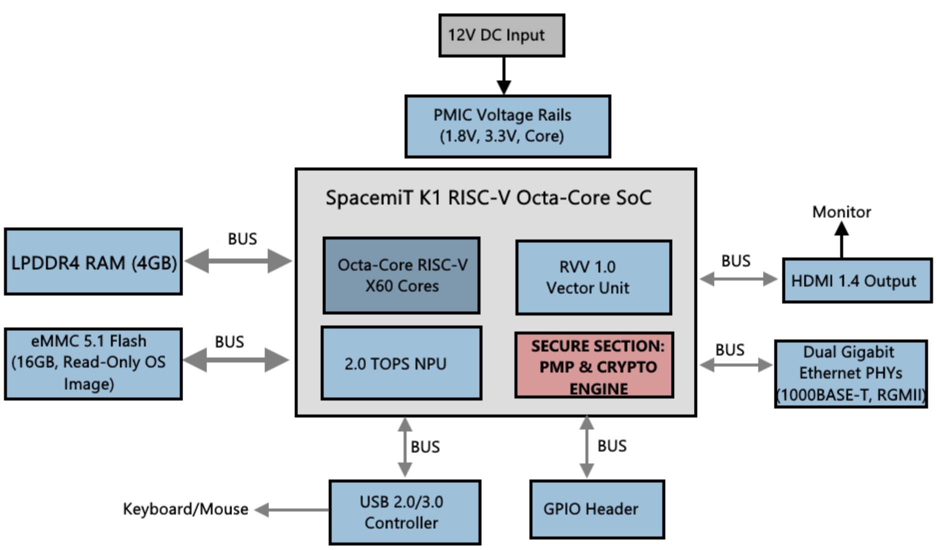


Figure 5: Hardware Draft Schematic: High Level SBC Schematic with Banana Pi

5 Stretch Goals

This design focuses on the core deliverables for the capstone sequence. The project is intended to continue as a faculty-led effort under Dr. Blanchard, with the longer-term aim of deployment in University of Florida courses for in-class exams, or as devices that could be issued or mailed to students taking exams remotely. As the project continues to evolve toward that end goal, there is ample room for stretch work; the following objectives represent directions the team may pursue if time and pace allow.

- **Configuration distribution system.** A centralized mechanism to push exam settings, network configuration, and SEB configuration to deployed devices. This would allow instructors or lab staff to update many units without reflashing or manual per-device setup, reducing operational overhead as the fleet grows.
- **Hardware-bound cryptographic keys.** Each device could store a unique private key (e.g., in secure storage or TPM-like hardware) used to decrypt or verify configuration and exam payloads. The exam environment would then run only on approved, registered hardware and could not be copied to an arbitrary machine, strengthening the chain of trust.
- **Additional hardening.** As the project matures, further hardening may be pursued: stricter kernel configuration (e.g., lockdown, reduced attack surface), capability-based restrictions in user space, and a stronger or more auditable secure boot chain. These would be prioritized based on risk assessment and time available after core functionality is complete.

References

[1] Safe Exam Browser. ETH Zürich. [Online]. Available: `https://github.com/safeexambrowser`. Accessed Mar. 2026.

[2] ETH Zürich, "Safe Exam Browser: Concept and Architecture," SafeExamBrowser.org, 2025. [Online]. Available: `https://safeexambrowser.org/about_overview_en.html`. Accessed Mar. 2026.

[3] SpacemiT, "SpacemiT K1 8 core RISC-V chip Brief," SpacemiT Technical Documentation, 2024. [Online]. Available: `https://docs.banana-pi.org/en/BPI-F3/SpacemiT_K1`.

[4] RISC-V International, "The RISC-V Instruction Set Manual, Volume II: Privileged Architecture," Document Version 20211203, Dec. 2021. [Online]. Available: `https://www.scs.stanford.edu/~zyedidia/docs/riscv/riscv-privileged.pdf`.

[5] J. Deitrick, "Cage: A Wayland kiosk compositor," GitHub Repository, 2025. [Online]. Available: `https://github.com/cage-kiosk/cage`.